

VISHNU VARDHAN MUTHA

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer and Analyst skilled in building end-to-end ELT pipelines, cloud data warehouses, and automated transformation layers. Currently completing an M.S. in Data Analytics at Webster University (USA). Hands-on experience designing a production-style data pipeline with Python, Snowflake, dbt, Apache Airflow, and Docker—ingesting live API data on an hourly schedule, transforming it with SQL models, and validating it with automated data-quality tests. Strong foundation in SQL, Python, dimensional/star-schema modeling, and data visualization. Seeking entry-level Data Engineer or Data Analyst roles in the USA.

TECHNICAL SKILLS

Data Engineering: ETL/ELT Pipelines, Apache Airflow, dbt, Snowflake, Data Warehousing, Star Schema & Dimensional Modeling, Data Quality Testing, Docker

Languages & Databases: Python (Pandas, NumPy, Scikit-learn), SQL, Java, PostgreSQL, Snowflake, PySpark (familiar)

Cloud & Tools: AWS, Docker, Git, GitHub, REST APIs, Jupyter Notebook, VS Code, PyCharm

BI & Visualization: Power BI, Tableau, Matplotlib, Seaborn, Excel

PROJECTS

AgentOps — AI Agent Analytics & Observability Platform — *Personal Project · 2026*

Python · AWS S3 · Snowflake · dbt · FastAPI · Streamlit · Airflow · Docker | [GitHub Repo](#)

- Built an end-to-end AI-agent observability platform processing 100,000 synthetic requests through a Bronze/Silver/Gold pipeline: AWS S3 data lake, Pandas validation, and Snowflake warehouse.
- Modeled a star schema in dbt (4 staging views, 4 fact/dimension tables) with 16 automated data-quality tests, and exposed the results via a FastAPI service with 8 REST endpoints for cost, latency, and agent-health analytics.
- Deployed both services independently on Render with Docker, and designed an hourly Apache Airflow DAG to orchestrate the full generate-to-warehouse pipeline.

Wildfire Risk & Air Quality Monitoring Pipeline — *Personal Project · 2026*

Python · Snowflake · dbt · Apache Airflow · Docker · REST API · SQL | [GitHub Repo](#)

- Engineered a production-style ELT pipeline that ingests live weather and air-quality data from the Open-Meteo REST API for 6 U.S. regions every hour, loading 42+ raw records per cycle into a Snowflake cloud data warehouse.
- Designed and built 3 dbt models (one staging plus two mart tables) in SQL—computing a 0–100 fire-risk score from temperature, humidity, and wind, and classifying PM2.5 readings into WHO/EPA health bands.
- Implemented 8 automated dbt data-quality tests (not-null, accepted-values) validating every output row, with all 8 passing to prevent bad data from reaching the analytics layer.
- Orchestrated the full ingest → dbt run → dbt test workflow with an Apache Airflow DAG on an hourly schedule, and containerized the entire stack with Docker for reproducible, environment-independent deployment.

Netflix Content Analysis — EDA & Insights — *Personal Project · 2025*

Python · Pandas · Matplotlib · Seaborn · Jupyter · Git | [GitHub Repo](#)

- Analyzed 8,800+ Netflix titles to surface content-acquisition trends, genre distribution, ratings, and catalog growth from 2008–2021 using Python and Pandas.
- Created 6 publication-ready visualizations identifying the USA as the top content producer and TV-MA as the dominant rating (36% of titles); published the full pipeline with clean GitHub documentation.

IPL Data Analysis — EDA & Visualizations — *Personal Project · 2026*

Python · Pandas · Matplotlib · Seaborn | [GitHub Repo](#)

- Performed end-to-end exploratory analysis on 13 IPL seasons (180,000+ ball-by-ball records) by joining 2 CSV datasets with Pandas—identifying top scorers, leading wicket-takers, and team win patterns.
- Produced 6 publication-ready visualizations covering season trends, toss decisions, and venue statistics; packaged with a standalone script, Jupyter notebook, and professional README.

EDUCATION

Master of Science, Data Analytics — Webster University, USA

Aug 2024 – Jun 2026

Bachelor of Technology, Computer Science — Samskruthi Engineering College, India

Graduated 2023

CERTIFICATIONS

- Google Data Analytics — Google / Coursera
- Introduction to Data Engineering — IBM / Coursera
- Cloud Onboarding for SAP Cloud ERP — SAP